

The $qx + 1$ Generalization of the Plancherel Framework

Nathan Humphrey

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Abstract

We ask whether the Plancherel-saturation structure that fixes the renormalized mass at $c = 7/45$ for the $3x + 1$ Syracuse map (Humphrey, 2026b) generalizes to the $qx + 1$ family for odd $q \geq 3$, and answer at three rigor tiers. First, the full-period Fourier transform $\widehat{F}_p$ satisfies a family-level magnitude formula $|\widehat{F}_p(\xi)| = p^{(r+3)/2}$ on a support of size p^{r-1} ; its residual equidistribution lemma, previously empirical at $q = 3$, is now established by a machine-checked second-moment identity formalized in Lean 4 and verified to machine precision at small p , leaving a single cited standard fact (a binomial lift) as the only undischarged hypothesis. Second, the short-window character sum K_p exhibits universal $\sqrt{N}$ saturation across $p \in \{3, \dots, 23\}$, though the bilinear bound that would explain it is not produced; we locate the Burgess-type wall precisely. Third, the renormalized mass follows a candidate $\tilde{c}_q = (q - 3)/q$ governed by a universal $q/3$ ratio, with $q = 5, 7$ deviations reported transparently. We state three negative claims upfront—chiefly that none of this dissolves the conditional dependence on Tao (2022) Proposition 1.17, since $\widehat{F}_p$ is the wrong object to substitute for $\widehat{\mu}_n$. The work was pre-registered.

1 Introduction

For odd integer n , the $qx + 1$ map sends $n \rightarrow (qn + 1)/2^v$ where 2^v is the largest power of 2 dividing $qn + 1$. At $q = 3$ this is the standard Syracuse compression of the Collatz map, whose long-time behavior was first analyzed via random-walk heuristic by Lagarias (1985) and surveyed comprehensively in Lagarias (2010). The associated Markov chain $\text{Syrac}(\mathbb{Z}/q^k\mathbb{Z})$ on $(\mathbb{Z}/q^k\mathbb{Z})^*$, introduced in Tao (2022) building on Terras (1976), has a stationary measure π_k whose characteristic function $\widehat{\mu}_k(\xi)$ encodes the Fourier-side structure of the chain. In a companion paper (Humphrey, 2026b) we proved

$$\lim_{k \rightarrow \infty} S_k^{(3)} = 7/15, \quad \text{where } S_k^{(q)} := \sum_{\substack{\xi \in \mathbb{Z}/q^k \\ q \nmid \xi}} |\widehat{\mu}_k(\xi)|^2, \quad (1)$$

and equivalently $c := \lim_{k \rightarrow \infty} \|d_k\|^2 \cdot 3^{k-1} = 7/45$, where d_k is the level- $(k-1)$ -to- k lift residual.

The present paper asks: does the structure underlying Equation 1 generalize to $qx + 1$? We answer at three rigor tiers.

1.1 Finding 1 (rigorous-pending).

The full-period magnitude formula generalizes. Define $f_p(u) := e_M(c \cdot (1+p)^u)$ for $u \in \mathbb{Z}/M$ with $M = p^{r+1}$, and let $\widehat{F}_p(\xi) := \sum_{u=0}^{M-1} e_M(c \cdot (1+p)^u - \xi u)$ be its full-period Fourier transform on $\mathbb{Z}/M$. The residual proof ingredient—principal-unit Gauss-sum equidistribution, left as an empirical observation at $q = 3$ in (Humphrey, 2026b)—is now established by a machine-checked second-moment identity $|G_p[a]|^2 = p^{r+1} \mathbf{1}[a \equiv c \pmod{p}]$, formalized in Lean 4 (**Mathlib**) for every odd prime $p \geq 3$ and $r \geq 1$. The formalization discharges the order and lifting-the-exponent inputs ($H1$, $H2$) from **Mathlib** and rests on a single cited standard fact ($H3$, the binomial lift $(1+p)^{p^{r-1}} \equiv 1 + p^r \pmod{p^{r+1}}$). The two remaining proof steps being p -blind, we state as a theorem candidate—closed modulo $H3$ —that for every odd prime $p \geq 3$ and $r \geq 2$,

$$|\widehat{F}_p(\xi)| = p^{(r+3)/2} \quad \text{for every } \xi \in \text{supp}(\widehat{F}_p), \quad (2)$$

with $|\text{supp}(\widehat{F}_p)| = p^{r-1}$. Equation (2) is verified to machine precision at $p \in \{3, 5, 7\}$, $r \in \{2, 3\}$, and the proof template (Cochrane’s Theorem 2 applied to the p -adic polynomial $g(u) = c \cdot (1+p)^u - p^2 mu$, combined with Plancherel and principal-unit Gauss-sum equidistribution) is p -blind.

1.2 Finding 2 (empirical).

The short-window character sum $K_p(r, c, m) := \sum_{u=0}^{N-1} e_M(c \cdot (1+p)^u - p^2 mu)$ with $N = p^{r-1} = M/p^2$ exhibits universal $\sqrt{N}$ saturation across $p \in \{3, 5, 7, 11, 13, 17, 19, 23\}$: under the canonical metric $|K_p(r, 1, 0)|/\sqrt{N}$ remains in the band $[0.83, 1.17]$ with regression slope $\beta_p \in [0.483, 0.518]$ on $\log |K|$ vs. $\log N$ (regression $R^2 \geq 0.996$). The prefactor C_p varies modestly ($\sim 1.4\times$ across the tested primes) with no apparently arithmetic correlate; rigorous closure of $\sqrt{N}$ saturation requires a Burgess-type bilinear bound on the arithmetic-progression character sum, the same wall encountered at $q = 3$ in our prior work.

1.3 Finding 3 (frontier-empirical).

The renormalized Plancherel mass ratio

$$\tilde{c}_q := \lim_{k \rightarrow \infty} \frac{S_k^{(q)}}{q/3^k} \quad (3)$$

exists for every prime q tested. The universal ratio $S_{k+1}^{(q)}/S_k^{(q)} \rightarrow q/3$ is empirically clean to four significant figures by $k = 2$ at $q \in \{11, 13\}$. The closed-form candidate

$$\tilde{c}_q \approx \frac{q-3}{q} \quad (4)$$

matches the data at $q \in \{11, 13, 17\}$ within 1%. The cases $q = 5$ ($\delta = 0.22$) and $q = 7$ ($\delta = 0.37$) deviate substantially from Equation 4; we do not, from currently available finite- k data, distinguish whether these deviations are finite- k transients converging to Equation 4 at higher k or genuine structural corrections.

1.4 Three negative claims, stated upfront.

The findings above admit no overstatement that should be permitted at outline level:

1. We do not prove $\tilde{c}_q = (q-3)/q$ rigorously for any q . The match at $q \in \{11, 13, 17\}$ is empirical.
2. We do not close the K_p short-window $\sqrt{N}$ saturation rigorously at any q . The structural argument that the wall has the same shape at every prime is rigorous; the bilinear bound that would jump the wall is open.
3. Finding 1.1 does not dissolve the conditional dependency of Humphrey (2026b)’s Theorem on Tao (2022) Proposition 1.17. The object $\widehat{F}_p$ is a deterministic character sum on $\mathbb{Z}/p^{r+1}$; Proposition 1.17 bounds the Markov-chain stationary’s characteristic function $\widehat{\mu}_n$. The two are distinct, and Finding 1’s $\widehat{F}_p$ magnitude is the wrong shape to substitute.

1.5 Position relative to the corpus.

”The Algebraic Content of Mod 8 Residue Structure in Collatz Stopping Times” (Humphrey, 2026a) established the mod 8 residue structure. ”A Symbolic Prefix Decomposition of the Collatz Map and Its Empirical Determination of Per Class Statistics” (Humphrey, 2026c) established $\langle \alpha_{\text{det}} \rangle = \log 6 / \log(4/3)$ as a theorem and verified the per-class moment clustering empirically at $k \in \{6, 7, 8, 9\}$. ”Closed-Form Plancherel Mass of the Syracuse Stationary Measure: The Constant 7/45” (Humphrey, 2026b) established $c = 7/45$ for $q = 3$. The present paper extends the framework of the third paper to general q at three rigor tiers, with $q = 3$ recovered as a special case (or, in the case of Equation 4, as a separate regime, see §5).

1.6 Organization.

Section 2 fixes the three objects $\widehat{F}_p, K_p, \widehat{\mu}_n$ and reviews how the Plancherel framework of Humphrey (2026b) extends in form to general q . Section 3 states Find 1.1 with its proof template and the residual equidistribution gap. Section 4 reports Finding 1.2 with the $p \in \{3, \dots, 23\}$ data and identifies the bilinear-bound wall. Section 5 presents Finding 1.3, including the honest treatment of $q = 5, 7$ deviations. Section 6 records that the prefix decomposition of Humphrey (2026c) formally extends to $qx + 1$ and reports the empirical degradation pattern of its predictive content. Section 7 states the pre-registration discipline under which the attempt was executed (commit f96fb86). Section 8 lists open questions and positions a follow-up paper tightening the Tao bridge. Section 9 concludes.

1.7 Adjacent contemporary work.

Chang (2026a) and Chang (2026b) analyze symbolic decomposition on the $a \cdot m + c$ state space and report parallel $q \equiv 7 \pmod{8}$ observations using a block total-variation decomposition of burst-gap indicators. The frameworks are distinct: their decomposition is on the burst-gap indicator of trajectory rather than on the $\widehat{F}_p/K_p/\widehat{\mu}_n$ objects of the present paper. We note the parallel mod-8 observation as a separate independent line of evidence consistent with the residue-class structure of Humphrey, 2026a.

2 Setup: three distinct Fourier objects

The reader is asked to hold three objects apart through the rest of the paper. They are related but not identifiable, and conflating them is the principal failure mode of generalization claims in this area.

2.1 Object A: the deterministic base function.

For an odd prime $p \geq 3$ and integer $r \geq 2$, set $M = p^{r+1}$. The principal units $1 + p\mathbb{Z}_l \subset \mathbb{Z}_l^*$ form a procyclic group; in particular $(1 + p)$ has multiplicative order p^r in $(\mathbb{Z}/M\mathbb{Z})^*$ (Neukirch, 1999). The function

$$f_p(u) := e_M(c \cdot (1 + p)^u), \quad u \in \mathbb{Z}, \quad (5)$$

where $e_M(x) := \exp(2\pi i x/M)$ and $c \in (\mathbb{Z}/M)^*$ is a fixed coefficient, is therefore periodic in u with period p^r . The cases $c \equiv 0 \pmod{p}$ are degenerate; we restrict to $c \not\equiv 0 \pmod{p}$ throughout.

2.2 Object B: the full-period Fourier transform $\widehat{F}_p$.

The full-period Fourier transform of f_p on $\mathbb{Z}/M$ is

$$\widehat{F}_p(\xi) := \sum_{u=0}^{M-1} e_M(c \cdot (1 + p)^u - \xi u), \quad \xi \in \mathbb{Z}/M. \quad (6)$$

By the period structure of f_p (period p^r , length $M = p \cdot p^r$), $\widehat{F}_p$ vanishes outside the sublattice $p \cdot \mathbb{Z}/M$. Within that sublattice, $\widehat{F}_p$ is further supported only on $\xi = p \cdot a$ for a in a specific subset $S_p^{(r)} \subseteq \mathbb{Z}/p^r$ determined by the principal-unit character decomposition; $|S_p^{(r)}| = p^{r-1}$ and the support shape is invariant under multiplication by $(\mathbb{Z}/p)^*$ within each principal-unit coset (Cochrane and Zheng, 2002).

2.3 Object C: the short-window character sum K_p .

For the same (p, r, c) and an additional parameter $m \in \mathbb{Z}/p$, define

$$K_p(r, c, m) := \sum_{u=0}^{N-1} e_M(c \cdot (1 + p)^u - p^2 m u), \quad N := p^{r-1}. \quad (7)$$

This is the truncation of the full-period sum to its first $N = M/p^2$ terms with a frequency shift by $p^2 m$. It is the object whose asymptotic magnitude controls a number of Plancherel-saturation arguments in the

literature (Kalafatelis et al., 2024), and is the object on which the empirical $\sqrt{N}$ behavior of Finding 1.2 is measured.

2.4 Object D: the Markov-chain stationary's characteristic function $\hat{\mu}_n$.

Let π_n denote the stationary distribution of $\text{Syrac}(\mathbb{Z}/q^n)^*$ (Tao, 2022). Define

$$\hat{\mu}_n(\xi) := \sum_{r \in (\mathbb{Z}/q^n)^*} \pi_n(r) \cdot e_{q^n}(-r\xi). \quad (8)$$

This is the object Proposition 1.17 of Tao (2022) bounds (under the Geometric(1/2) heuristic for 2-adic valuations of $3r + 1$ values).

2.5 The four objects are not the same.

$\hat{F}_p$ (Object B) is a deterministic character sum on a cyclic group; K_p (Object C) is its short-window truncations; $\hat{\mu}_n$ (Object D) is the Fourier transform of a probability measure on $(\mathbb{Z}/q^n)^*$. The relations are:

- K_p is recovered from $\hat{F}_p$ by Pólya-Vinogradov decomposition $K_p = M^{-1} \sum_{\xi \in \text{supp}(\hat{F}_p)} \hat{1}_N(\xi) \hat{F}_p(\xi)$, where $\hat{1}_N$ is the truncated indicator's Fourier transform. The relation is exact; the question is what magnitude bound on $\hat{F}_p$ implies what magnitude bound on K_p .
- The Cauchy-Schwarz bound on this decomposition, using only $|\hat{F}_p(\xi)|^2 \leq Mp^2$ and $\sum |\hat{1}_N|^2 \leq NM$, yields $|K_p| \lesssim p\sqrt{N}$ – the trivial bound. The $\sqrt{N}$ saturation (without the factor p) requires phase cancellation in the bilinear sum $\sum \hat{1}_N(\xi) \hat{F}_p(\xi)$; this is the Burgess-type bilinear question.
- $\hat{\mu}_n$ is unrelated to $\hat{F}_p$ and K_p at the level of Object identity. There is no extant computation in the project that expresses $\hat{\mu}_n(\xi)$ as a deterministic character sum on a cyclic group. (Tao's Proposition 1.17 obtains the polynomial decay $|\hat{\mu}_n(\xi)| \ll_A n^{-A}$ via the trajectory expansion $\text{Syrac}(\mathbb{Z}/3^n) = 2^{-a_1} + 3 \cdot 2^{-a_{[1,2]}} + \dots$ with i.i.d. Geom(2) exponents, which is a Markov-chain rather than character-sum argument.)

2.6 The Plancherel framework at general q .

Three structural results of Humphrey (2026b) extend to general q in their statements, with the rigor of each extension differing:

- **H75_q** (Plancherel decomposition on $\mathbb{Z}/q^k$): $S_k^{(q)} = \sum_{\xi \in q^k \mathbb{Z}} |\hat{\mu}_k(\xi)|^2$. This is immediate from Plancherel on cyclic groups and holds for every q without further assumption.
- **H76_q** (Conservation law/leading-mode identity): the conservation identity $\sum_{j=0}^{q-1} M_{n+1}(\eta_0 + j \cdot q^n) = 0$ and the leading-mode reduction $S_{n+1} = -2M_{n+1}(1 + q^n)$ of Humphrey, 2026b, Theorems 76.1 and 76.3, are derived from the q -adic class structure of $(\mathbb{Z}/q^n)^*$ and the Markov-chain transition kernel. The class-symmetry chain rule $K_- = \sigma K_+ \sigma$ extends to every odd prime q via the analogous involution on $(\mathbb{Z}/q^n)^*$. The full proofs at general q are not given in the present paper; they replicate Humphrey (2026b)'s §76 with q in place of 3, and we treat the extensions as available rather than proven here.
- **H77_q** (T-analog): Humphrey (2026b) Theorem 77.1 gives the diagonal-only contribution of Tao's bilinear recursions at $q = 3$ as $T_{\text{diag}} = (1/5) \cdot [[1, 1], [4, 4]]$, with characteristic polynomial $\lambda^2 - \lambda$, eigenvalue 1 on $(1, 4)$, eigenvalue 0 on $(1, -1)$. The natural generalization is a 2×2 matrix at each q whose $(1, a_q)$ -eigenvector ratio is fixed by the squared class-mass ratio at the $(\mathbb{Z}/3)^*$ level – at $q = 3$ this is $(1/3)^2 : (2/3)^2 = 1 : 4$, giving the $a_3 = 4$ that produces the eigenvector $(1, 4)$. At general q the analogous ratio is open: while the class-balance argument of Humphrey, 2026b §H64.B uses only the asymptotic class distribution is $(0, 1/3, 2/3) \bmod 3$, the analogous distribution $\bmod q$ for general q is not established in the present work. We state this as an open question rather than asserting an extension.

The asymmetry in extension rigor across $H75_q$, $H76_q$, $H77_q$ is real: Plancherel is automatic, the conservation law transfers cleanly, the T_{diag} analog is open. Section 5 confronts this honestly: the $\tilde{c}_q = (q - 3)/q$ closed form is empirical at the level of the renormalized Plancherel mass, not derived from a T_{diag} -analog spectrum at general q .

3 Theorem candidate: the $\widehat{F}_p$ magnitude formula

3.1 Statement

Theorem Candidate 3.1 ($qx + 1$ Plancherel saturation at the $\widehat{F}_p$ level). *For every odd prime $p \geq 3$ and integer $r \geq 2$, let $M = p^{r+1}$, $f_p(u) = e_M(c(1+p)^u)$ for $c \in (\mathbb{Z}/M)^*$, and $\widehat{F}_p(\xi)$ its full-period Fourier transform on $\mathbb{Z}/M$ as in Equation 6. Then*

i $\widehat{F}_p$ is supported on $\{p \cdot a : a \in S_p^{(r)}\}$ for a subset $S_p^{(r)} \subseteq \mathbb{Z}/p^r$ of size $|S_p^{(r)}| = p^{r-1}$ determined by the principal-unit character decomposition;

ii on its support, $\widehat{F}_p$ has uniform magnitude $|\widehat{F}_p(\xi)| = p^{(r+3)/2}$.

We name this a theorem candidate rather than a theorem because the third proof ingredient – principal-unit Gauss-sum equidistribution – now rests on a single cited standard fact (H3) rather than on a closed-form derivation from first principles. At $q = 3$, (Humphrey, 2026b) verified the equidistribution empirically. We have since established it at general p by a machine-checked second-moment computation: the identity $|G_p[a]|^2 = p^{r+1} \mathbf{1}[a \equiv c \pmod{p}]$ is formalized in Lean 4 for every odd prime $p \geq 3$ and $r \geq 1$, with the order (H1) and lifting-the-exponent (H2) inputs discharged from `Mathlib` and the binomial lift (H3) supplied as a cited hypothesis. Promotion to a full theorem awaits only the formal discharge of H3, which is standard (Neukirch, 1999, Ch. 4); the gap, formerly an unproven equidistribution property at general p , is now a single textbook lemma.

3.2 Proof template (p -blind)

The proof is the family-level extension of Humphrey (2026b) §78.1-78.3. We sketch the three steps and identify what is p -blind versus what needs the equidistribution lemma.

Step 1: Sparsity from period structure. By construction, $(1+p)$ has order p^r in $(\mathbb{Z}/M)^*$ since $M = p^{r+1}$ and the principal-unit subgroup $1 + p\mathbb{Z}/M$ is cyclic of order p^r (Neukirch, 1999). Therefore f_p is periodic in u with period p^r , and its Fourier transform on $\mathbb{Z}/M$ vanishes outside the sublattice $p \cdot \mathbb{Z}/M$, which has size $M/p = p^r$. This step uses only the standard fact that the Fourier transform of a T -periodic function on a group order $T \cdot s$ is supported on a sublattice of index T . It is p -blind.

Step 2: Further sparsity via principal-unit decomposition. Restricting to $\xi = p \cdot a$ with $a \in \mathbb{Z}/p^r$, the sum of Equation 6 factors through the principal-unit character group. The characters of $1 + p\mathbb{Z}/M$ that are nontrivial on the full group correspond to $a \in (\mathbb{Z}/p^r)^*$; among those, only the $|S_p^{(r)}| = p^{r-1}$ characters whose restriction to the smallest principal-unit subgroup is nontrivial actually contribute to Equation 6 with $c \not\equiv 0 \pmod{p}$. (The remaining characters give complete-sum vanishing via the Cochrane Theorem 2 argument applied to the p -adic polynomial $g(u) = c(1+p)^u - \xi u$, which has p -adic degree determined by the principal-unit structure rather than p itself.) This step uses Cochrane’s Theorem 2 (Cochrane and Zheng, 2002) as the structural input; its p -blindness follows from the fact that Cochrane’s argument depends on the p -adic degree, not on p itself.

Step 3: Magnitude via Plancherel and equidistribution. By Plancherel on $\mathbb{Z}/M$,

$$\sum_{\xi \in \mathbb{Z}/M} |\widehat{F}_p(\xi)|^2 = M \cdot \sum_{u=0}^{M-1} |f_p(u)|^2 = M^2. \quad (9)$$

Restricting to the support (size p^{r-1}),

$$\sum_{\xi \in \text{supp}(\widehat{F}_p)} |\widehat{F}_p(\xi)|^2 = M^2, \quad (10)$$

since $\widehat{F}_p$ vanishes outside the support. If the magnitudes on the support are uniform – the equidistribution property – then each term equals $M^2/p^{r-1} = M \cdot p^2 = p^{r+3}$, so $|\widehat{F}_p(\xi)| = p^{(r+3)/2}$ on the support. The equidistribution property is the residual step. It is known structurally to follow from principal-unit Gauss-sum theory (the magnitude of a Gauss sum on a primitive character of a cyclic group of order p^r is exactly $\sqrt{p^r}$ (Ireland and Rosen, 1990)), but the project’s prior corpus did not produce a closed-form derivation of equidistribution valid at general p ; the $q = 3$ instance was verified empirically.

3.3 Empirical Verification

Table 1 reports $|\widehat{F}_p(\xi)|$ at representative support points for $p \in \{3, 5, 7\}$, $r \in \{2, 3\}$. The predicted magnitude $p^{(r+3)/2}$ matches the computed value to machine precision (relative error $< 10^{-12}$ at every cell).

Table 1: Empirical $|\widehat{F}_p|$ magnitude versus the predicted $p^{(r+3)/2}$.

p	r	$M = p^{r+1}$	predicted $p^{(r+3)/2}$	observed	
3	2	27	$\sqrt{27} \approx 5.1962$	5.1962	(exact)
3	3	81	$\sqrt{81} = 9$	9.0000	(exact)
5	2	125	$\sqrt{125} \approx 11.1803$	11.1803	(exact)
5	3	625	$\sqrt{625} = 25$	25.0000	(exact)
7	2	343	$\sqrt{343} \approx 18.5203$	18.5203	(exact)
7	3	2401	$\sqrt{2401} = 49$	49.0000	(exact)

3.4 The residual rigor gap, now closed in Lean modulo one cited fact

The formerly unsettled step was the equidistribution of $|\widehat{F}_p|$ on its support: whether the principal-unit Gauss sums have constant magnitude across the support. We have closed this step by a machine-checked computation. Formalized in Lean 4 (`Mathlib`, file `CollatzVerify/FhatMagnitude.lean`), the second-moment identity

$$|G_p[a]|^2 = p^{r+1} \mathbf{1}[a \equiv c \pmod{p}] \quad (11)$$

holds for every odd prime $p \geq 3$ and $r \geq 1$. It exhibits both the support ($a \equiv c \pmod{p}$) and the uniform magnitude $\sqrt{p^{r+1}}$ directly, without an assumed equidistribution hypothesis: support and constancy fall out of the same computation. The proof discharges the order of $1+p$ ($H1$) and the lifting-the-exponent valuation ($H2$) from `Mathlib`, and cites one standard fact ($H3$, the binomial lift $(1+p)^{p^{r-1}} \equiv 1 + p^r \pmod{p^{r+1}}$, e.g. (Neukirch, 1999, Ch. 4)) as a hypothesis. The remaining work to elevate Theorem Candidate 3.1 to an unconditional theorem is the formal discharge of $H3$, which is textbook and p -blind. We therefore report the equidistribution gap as closed modulo a single cited lemma, not as an open empirical observation.

4 Empirical: K_p universal $\sqrt{N}$ saturation

4.1 Object and result

For the short-window sum $K_p(r, c, m)$ defined in Equation 7, we study the dependence of $|K_p|$ on $N = p^{r-1}$ across primes. Define the canonical metric

$$|K_p^{c=1, m=0}(r)| = \left| \sum_{u=0}^{N-1} e_M((1+p)^u) \right| \quad (12)$$

and the saturation prefactor $C_p(r) = |K_p^{c=1, m=0}(r)|/\sqrt{N}$. The main empirical observation:

Empirical Observation 4.1 (Universal $\sqrt{N}$ saturation). *For $p \in \{3, 5, 7, 11, 13, 17, 19, 23\}$ and the r -range tested at each prime (Table 2), the canonical-metric prefactor $C_p(r)$ remains in the band $[0.83, 1.17]$. Fitting $\log |K_p^{c=1, m=0}| = a + \beta_p \log N$ by ordinary least squares at each prime yields slopes $\beta_p \in [0.483, 0.518]$ with $R^2 \geq 0.996$ across all tested primes. We do not claim that $K_p/\sqrt{N}$ converges to a constant as $r \rightarrow \infty$; we report that it remains in the cited band over the tested r -window.*

Table 2: Empirical $|K_p|$ saturation across primes (canonical metric $c = 1$, $m = 0$).

p	r -range tested	OLS slope β_p	R^2	C_p mean
3	8–20	0.494	0.997	0.88
5	6–12	0.483	0.999	0.94
7	6–10	0.484	0.998	0.98
11	5–8	0.517	0.996	1.06
13	4–7	0.518	0.997	1.00
17	3–4	—	—	1.02
19	3–4	—	—	0.94
23	3–4	—	—	1.03

For $p \in \{17, 19, 23\}$, the available r -window is too narrow to fit a slope reliably; we report only the mean prefactor and note that all observed $|K_p|/\sqrt{N}$ values across these primes fall in $[0.93, 1.10]$, consistent with the band observed at smaller primes.

4.2 The structural-versus-empirical separation

Observation 4.1 is empirical. The reasoning that explains why $\sqrt{N}$ is the natural rate – the Pólya-Vinogradov decomposition of K_p in terms of $\widehat{F}_p$ – is p -blind in structure: at every prime, the decomposition takes the same shape, the Cauchy-Schwarz bound on it gives the trivial $\sqrt{N} \cdot p$ scaling, and the $\sqrt{N}$ saturation (without the extra p) requires the same bilinear-bound step at every prime. But the rigorous bilinear bound is not produced here at any prime.

Concretely: writing $K_p = M^{-1} \sum_{\xi} \widehat{1}_N(\xi) \widehat{F}_p(\xi)$ with $\widehat{1}_N(\xi) = \sum_{u=0}^{N-1} e_M(-\xi u)$, the Cauchy-Schwarz bound gives

$$|K_p|^2 \leq \frac{1}{M^2} \cdot |\text{supp}(\widehat{F}_p)| \cdot \max_{\xi} |\widehat{F}_p(\xi)|^2 \cdot \sum_{\xi} |\widehat{1}_N(\xi)|^2. \quad (13)$$

Using $|\widehat{F}_p(\xi)|^2 = Mp^2$ from Theorem Candidate 3.1, $|\text{supp}(\widehat{F}_p)| = M/p^2 = N$, and $\sum |\widehat{1}_N|^2 \leq NM$ (Plancherel for the truncated indicator), this gives $|K_p|^2 \leq N^2 p^2$, hence $|K_p| \leq p\sqrt{N}$. This is the trivial bound or worse.

The observed saturation $|K_p| \approx C_p \sqrt{N}$ with $C_p = O(1)$ requires phase cancellation in the bilinear sum $\sum \widehat{1}_N(\xi) \widehat{F}_p(\xi)$ across the support of $\widehat{F}_p$. This is a Burgess-type bound on a character sum restricted to an arithmetic-progression-like coset, and it is the same wall that prevents rigorous closure of $\sqrt{N}$ saturation at $q = 3$ in our prior work.

4.3 What the prefactor variation means

The prefactor C_p varies by approximately a factor of 1.4 across the tested primes (Table 2, with $C_3 \approx 0.83$ at the largest r for $p = 3$ and $C_{13} \approx 1.17$). This variation has no obvious arithmetic correlate: C_p is not monotone in p , not correlated with $p \bmod 4$, not correlated with the multiplicative order of 2 mod p . We interpret this variation as living in the gap between $\widehat{F}_p$ (rigorous at the magnitude level via Theorem Candidate 3.1) and K_p (truncated, requiring the bilinear bound) – it is the residual of the phase-cancellation question, not a feature of a different magnitude law per prime. We do not have a structural account of the specific $1.4\times$ value.

5 The $\tilde{c}_q = (q - 3/q)$ candidate

We now turn to Finding 1.3, which concerns the renormalized Plancherel mass and is distinct in object-type from the character-sum work of §§3-4.

5.1 The universal $q/3$ ratio

Empirical Observation 5.1 (Universal $q/3$ growth ratio). *For every odd prime q tested, the level-to-level ratio of Plancherel mass satisfies $\frac{S_k^{(q)}}{S_k^{(q)}} \rightarrow \frac{q}{3}$ as $k \rightarrow \infty$. Empirically: the ratio is clean to four significant figures by $k = 2$ at $q \in \{11, 13\}$. At smaller primes the ratio settles less rapidly but trends to $q/3$ in every case tested.*

This ratio explains why the unrenormalized sequence $S_k^{(q)}$ diverges for $q \geq 5$ (since $q/3 > 1$) while $S_k^{(3)} \rightarrow 7/15$ converges (since $q/3 = 1$). The natural renormalization is therefore $\tilde{S}_K^{(q)} := S_k^{(q)}/(q/3)^k$, with $\tilde{c}_q := \lim_k \tilde{S}_K^{(q)}$.

5.2 The closed-form candidate

Empirical Observation 5.2 ($\tilde{c}_q$ values at the deepest tested k). *The renormalized constant $\tilde{c}_q$ is reported in Table 3, with the implied integer comparison $\tilde{c}_q \cdot q$ relative to the candidate value $q - 3$.*

Table 3: Empirical $\tilde{c}_q$ at the deepest tested k , with comparison to the $(q - 3)/q$ candidate.

q	$k_{\max}$	$\tilde{c}_q$ (observed)	$\tilde{c}_q \cdot q$	$q - 3$	$ \tilde{c}_q \cdot q - (q - 3) $
3	4	0.464 (unrenormalized 7/15)	1.39	0	special regime, see below
5	4	0.488	2.44	2	0.44
7	3	0.783	5.48	4	1.48
11	3	0.729	8.017	8	0.017
13	2	0.770	10.007	10	0.007
17	2	0.825 (approx)	14.03	14	0.03

At $q \in \{11, 13, 17\}$ the integer match to $q - 3$ is within 1% (and within 0.2% at $q \in \{11, 13\}$). The candidate closed from $\tilde{c}_q = (q - 3)/q$, equivalently $\tilde{c}_q \cdot q \in \mathbb{Z}$ with value $q - 3$, fits these three prime to a precision consistent with the level of k resolved.

The $q = 3$ regime. At $q = 3$, $(q - 3)/q = 0$ identically, but the unrenormalized $S_k^{(3)} \rightarrow 7/15$. The candidate formula vanishes at $q = 3$ because $q/3 = 1$ means no renormalization is required; the limit $\tilde{c}_3 = 7/15$ is the unrenormalized Tao limit of Humphrey, 2026b, not the renormalized $(q - 3)/q$ value. This is consistent: at $q = 3$ the sequence $S_k^{(3)}$ converges to $7/15$ as $k \rightarrow \infty$, and within this convergent sequence the additional structure analyzed by $\tilde{c}$ for $q \geq 5$ is invisible.

5.3 Honest treatment of $q = 5$ and $q = 7$

The cases $q = 5$ and $q = 7$ do not fit the $(q - 3)/q$ candidate at the tested precision. The deviations are $\delta_5 = 0.22$ (relative deviation of $\tilde{c}_5 \cdot 5 = 2.44$ from $q - 3 = 2$) and $\delta_7 = 0.37$. We name two readings, both honest, and decline to pick between them on currently available data.

Reading A: finite- k transient. The tested k resolution is $k_{\max} = 4$ at $q = 5$ and $k_{\max} = 3$ at $q = 7$. The signal $\tilde{c}_q \cdot q \rightarrow q - 3$ may simply not have settled at these resolutions, with deeper k converging to the candidate value. Aitken extrapolation on the $k = 1, 2, 3, 4$ sequence at $q = 5$ suggests a limit near 0.485, which would deepen the deviation from $(q - 3)/q = 0.4$ rather than close it – though Aitken extrapolation is itself unreliable at only 4 points and on a sequence whose monotonicity is not established.

Reading B: structural correction not yet identified. A specific candidate is that the structural mechanism behind $(q-3)/q$ involves the multiplicative order of 2 mod q . Among the tested primes, $q = 7$ is the only case where 2 is not a primitive root modulo q (multiplicative order 3, not $6 = q - 1$); $q = 5, 11, 13$ all have 2 as a primitive root. If non-primitive-root status is the relevant distinguisher, $q = 17$ within 0.2% of the $(q-3)/q$ candidate, disfavoring this specific reading. We have not identified a structural mechanism that picks out $q = 5$ and $q = 7$ jointly while leaving $q = 11, 13, 17$ at clean match.

What we report. The cleanest finding is the $(q-3)/q$ match at $q \in \{11, 13, 17\}$. The deviations at $q = 5, 7$ are reported equal-prominence; we do not have a published claim that resolves them; the resolution requires either deeper k resolution (compute-bounded) or a structural derivation that picks out the deviations from first principles (open).

5.4 What $(q-3)/q$ would mean structurally

Were $\tilde{c}_q = (q-3)/q$ to hold rigorously, it would imply that $\tilde{c}_q \cdot q$ is an integer for every prime $q \geq 5$, with explicit value $q-3$. This integer-valued behavior across primes is itself the natural target for a structural derivation: it suggests an underlying combinatorial mechanism (a counting argument on q -adic class structure, perhaps counting "off-diagonal classes" in some appropriate sense) that produces a discrete count $q-3$ as the renormalized Plancherel mass. We do not produce such a derivation here. Note that the candidate formula's value at $q = 3$ is 0, which is consistent in the renormalized-mass sense (no off-diagonal structure visible after renormalization at $q = 3$ because no renormalization is needed) but obscures the unrenormalized $\tilde{c}_3 = 7/15$ results; the appropriate framing is that $(q-3)/q$ is a candidate for the renormalized mass on the $q \geq 5$ regime and that $q = 3$ is a seaparet base case with its own constant $7/15$ establish in Humphrey (2026b).

6 The prefix decomposition extends formally

The prefix decomposition of Humphrey (2026c) extends to $qx + 1$ in its algebraic content. We record the extensions and the empirical degradation pattern without claiming the empirical regularities transfer.

6.1 Algebraic extension

For odd n with $r = n \bmod 2^k$, with $n = 2^k m + r$ and track the symbolic state $a_t m + c_t$ through $qx + 1$ iteration starting from $(a_0, c_0) = (2^k, r)$. The parity rules generalize directly: if a_t is even and c_t is even $(a_{t+1}, c_{t+1}) = (a_t/2, c_t/2)$; if a_t is even and c_t is odd, $(a_{t+1}, c_{t+1}) = (qa_t, qc_t + 1)$; if a_t is odd, the prefix terminates.

Lemma 6.1 (Prefix determinism, $qx + 1$). *The prefix iteration starting from $(a_0, c_0) = (2^k, r)$ with r odd is fully determined by r and is independent of m .*

The proof replicates Humphrey (2026c) Lemma 1 with q in place of 3.

Theorem 6.2 (Termination, $qx + 1$). *The prefix iteration terminates in finitely many steps. The number of odd-step applications consumed during the prefix is at most k . The terminal a_{final} is of the form q^j for some integer j with $1 \leq j \leq k$.*

The proof replicates Humphrey (2026c) Theorem 1: $v_2(a_t)$ is monotone non-increasing along the prefix (even-step decrements by exactly 1; odd-step multiplies a by q , which is odd, so v_2 is unchanged), and the prefix terminates when $v_2(a_t) = 0$, with terminal $a_{\text{final}} = q^j$ where j counts the odd-step applications. The lower bound $j \geq 1$ uses that $c_0 = r$ is odd. The decomposition is a constructive sharpening of Terras (1976) Lemma 4 in the $qx + 1$ case, with q in place of 3: where Terras gives the asymptotic ratio between successive iterates' residue-class structure, the prefix decomposition retains the symbolic state through the deterministic prefix and exposes the terminal value $a_{\text{final}} = q^j$ as a closed-form covariate at finite k .

6.2 Empirical degradation across q

The empirical regularities of Humphrey (2026a) §4 transfer in form but degrade quantitatively. Table 4 summarizes; the underlying data is from `qx.plus_1.framework.csv` in the supplementary repository.

Table 4: Empirical degradation of prefix-decomposition regularities across $qx + 1$ at fixed modular resolution $k = 6$.

q	per-class slope CV	$\langle K_h \rangle$ closed-form vs. empirical gap	sample size at $N = 10^8$
3	3.7%	−2.1%	500,000
5	24.5%	+31.9%	32,785
7	— (sparse)	+13.6%	258
9	— (too sparse)	—	104

Three points worth naming:

- i The per-class slope variability grows from 3.7% at $q = 3$ to 24.5% at $q = 5$. The “slopes are universal across classes” property of Humphrey (2026a) §4 does not extend cleanly to $q = 5$. Possible mechanisms include class-dependence of the Cramér-tilted Geom(1/2) distribution under conditioning on convergence to the trivial cycle, or structural differences in the absorbing-class set $\{m_j(q)\}$ across q . The q -specific cycle structure was cataloged in Lagarias (1990): for $q = 5$ there exist nontrivial cycles of length 13 and 17 in addition to the trivial cycle; for $q = 7$ no nontrivial cycle below the tested bound is known, and similarly for $q = 9$.
- ii The Cramér multiplier $C \approx 5/2$ in the convergence rate $\text{conv_rate}(j; q) \approx A(q) \cdot (4/q)^{C \cdot j}$ across $q \in \{5, 7, 9\}$ is empirically clean ($R^2 \geq 0.99$ in all three cases). We do not derive the value $5/2$ here.
- iii The conditional-on-convergence functional form $K_h(q; \text{conv}) = (1 + \mathbb{E}^*[v]) / (\mathbb{E}^*[v] \log 2 - \log q)$ matches empirical pooled slopes within 2% at $q = 3$, degrading to 14% gap at $q = 7$. The structural form is universal; the quantitative match is q -dependent.

We name these as observations rather than theorems. The prefix decomposition’s empirical content at $q \neq 3$ is open, and deserves a dedicated treatment beyond the scope of the present paper.

7 Pre-registration discipline

The work reported in §§ 3-5 was executed under a pre-registration locked a commit `f96fb86` on 2026-05-11 before any compute was run. The pre-registration specified five disposition labels in advance:

1. **THEOREM_PROVEN** - family-level rate-1/2 universality proven rigorously, dissolving the conditional in Humphrey (2026b).
2. **CLAIMS_1_AND_2** - both the magnitude formula and the prefactor closed form rigorous.
3. **CLAIM_1_ONLY** - magnitude formula rigorous; prefactor empirical.
4. **EMPIRICAL_UNIVERSAL_NOT_STRUCTURAL** - $\sqrt{N}$ saturation universal across q but no structural argument.
5. **STRUCTURAL_OBSTRUCTION_FOUND** - a specific obstruction to the original framing is identified.

The outcome landed at **CLAIM_1_PARTIAL + STRUCTURAL_OBSTRUCTION_FOUND**: Theorem Candidate 3.1 closes structurally at family level (with the residual equidistribution lemma), but the original framing – that family-level rate-1/2 universality would dissolve Humphrey (2026b)’s conditional dependency on Tao Proposition 1.17 – is reported as unsuccessful. The object $\hat{F}_p$ is, as discussed in § 2, the wrong shape to substitute for $\hat{\mu}_n$.

We report this walk-back at equal prominence with the positive findings. It is a feature of the pre-registration framework that the original strategic goal can fail while substantive new findings emerge; the present paper exemplifies this pattern.

8 Open questions

We isolate three open questions whose resolution would substantively advance the $qx + 1$ chapter of the corpus.

8.1 Formal discharge of the binomial lift ($H3$)

The equidistribution lemma at general p is now machine-verified in Lean 4 (§3), reducing the residual rigor gap in Theorem Candidate 3.1 to the formal discharge of a single cited fact: the binomial lift $H3$, $(1 + p)^{p^{r-1}} \equiv 1 + p^r \pmod{p^{r+1}}$ for odd p . This is standard ((Neukirch, 1999, Ch. 4)); the formalization currently carries it as a hypothesis. Discharging $H3$ from `Mathlib`—a cofactor-tracking induction on the binomial expansion—would render Theorem Candidate 3.1 an unconditional, fully machine-checked theorem at family level. We flag this as the single remaining formal step.

8.2 The Burgess-type bilinear bound on K_p

Rigorously closing $|K_p| \leq C\sqrt{N}$ at any prime, with C uniform in p , requires a Burgess-type bilinear bound on the arithmetic-progression character sum that arises in the Pólya-Vinogradov decomposition of K_p . Burgess (1957) established the prototype bound for short character sums; the question is whether a family-level analog tailored to the principal-unit subgroup structure can be established. We pose this as the central open question for closing Observation 4.1.

8.3 Rigorous derivation of $(q - 3)/q$ add explanation of $q = 5, 7$ deviations

The closed-form candidate $\tilde{c}_q = (q - 3)/q$ at $q \in \{11, 13, 17\}$ asks for a structural derivation. The integer-valued behavior $\tilde{c}_q \cdot q = q - 3$ suggests a counting argument on q -adic class structure that we have not identified. Honestly reporting the $q = 5, 7$ deviations equal-prominence with the $q \in \{11, 13, 17\}$ match, the question is whether (a) the deviations are finit- k transients (resolvable by deeper compute) or (b) genuine structural corrections (resolvable only by a derivation that picks out their structural cause).

8.4 Position of a follow-up paper

A natural follow-up is a paper tightening the Tao bridge between Humphrey (2026c)’s $\alpha_{\det}$ predictor and Tao (2022)’s almost-bounded-values theorem at finite N . Empirical work in our prior corpus (a 40-cell verification at slope 1.000 ± 0.005 across $N \in \{2^{25}, 2^{27}, 2^{28}, 2^{30}, 2^{32}\}$, with the structural offset $\bar{\sigma} - K_h \log N \approx -2.45$ stable within 0.01 across 7 doublings of N) supports a quantitative bridge equation $s_{\text{mean}}(r) \approx \alpha_{\det}(r) + K_h \log(N/f(N)) + \varepsilon$. We defer this to subsequent work.

9 Conclusions

We have presented three findings extending the Plancherel framework of Humphrey (2026b) from $q = 3$ Syracuse to general $qx + 1$: a rigorous-pending magnitude formula $|\widehat{F}_p(\xi)| = p^{(r+3)/2}$ (Theorem Candidate 3.1), an empirical universal $\sqrt{N}$ saturation of the short-window sum K_p across $p \in \{3, \dots, 23\}$ (Observation 4.1), and a frontier-empirical closed-form candidate $\tilde{c}_q = (q - 3)/q$ matched within 1% at $q \in \{11, 13, 17\}$ (§ 5).

The $c = 7/45$ constant of Humphrey (2026b) for $q = 3$ Syracuse is the special case of a broader family: the universal $q/3$ ratio (Observation 5.1) is the structural backbone connecting the iteration levels at every prime, and the renormalized constant $\tilde{c}_q$ exists at every tested q . The cleanest signal is the integer match $\tilde{c}_q \cdot q = q - 3$ at $q \in \{11, 13, 17\}$.

We have stated three negative claims at equal prominence with the positive findings: the closed form is not rigorous at any q , the K_p saturation is not rigorous at any q , and the $\widehat{F}_p$ magnitude formula does not dissolve the conditional dependency of Humphrey (2026b) on Tao (2022)’s Proposition 1.17. The corpus advances at three rigor tiers rather than uniformly; we have aimed to report which finding lives at which tier with the precision the corpus discipline requires.

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